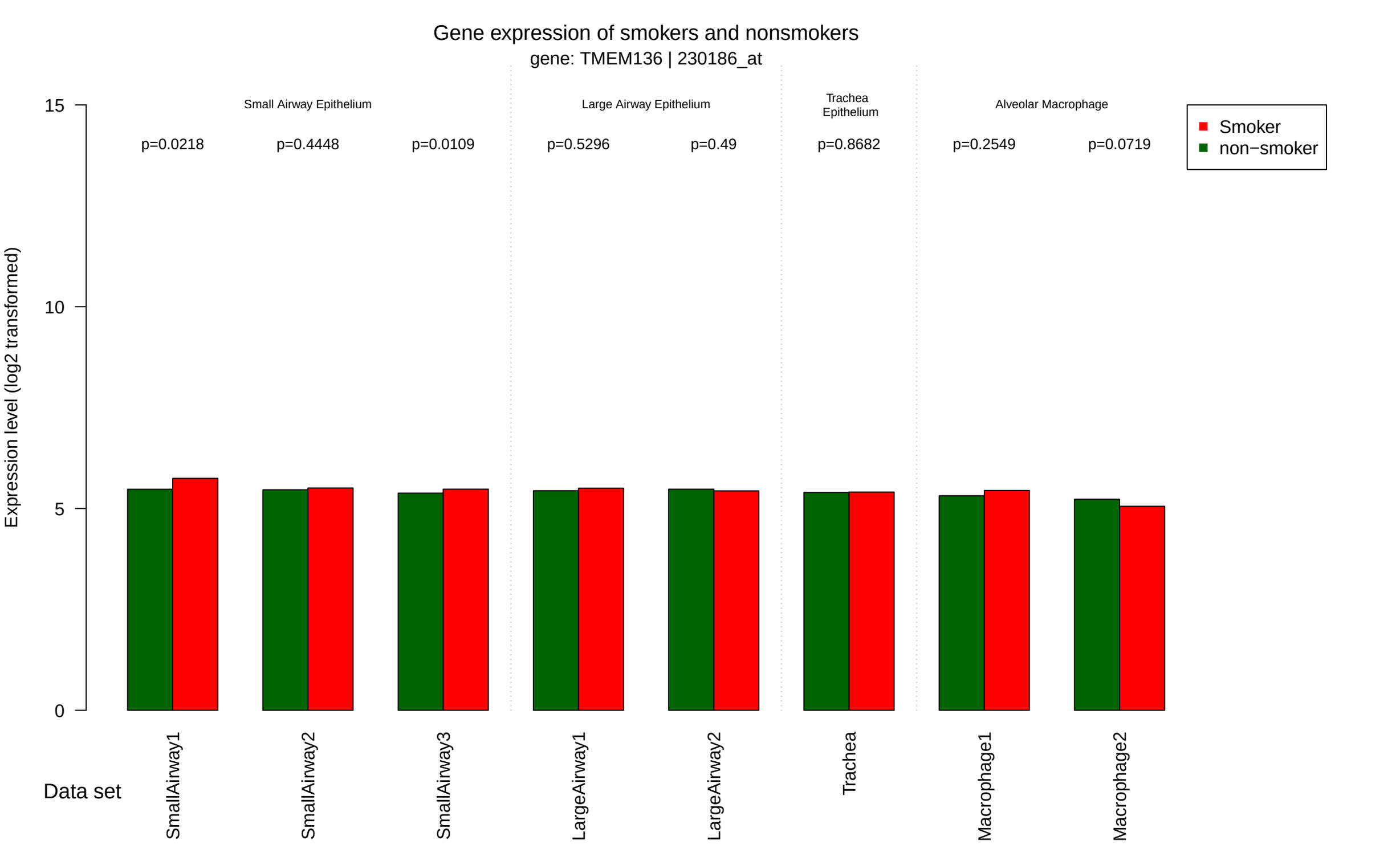
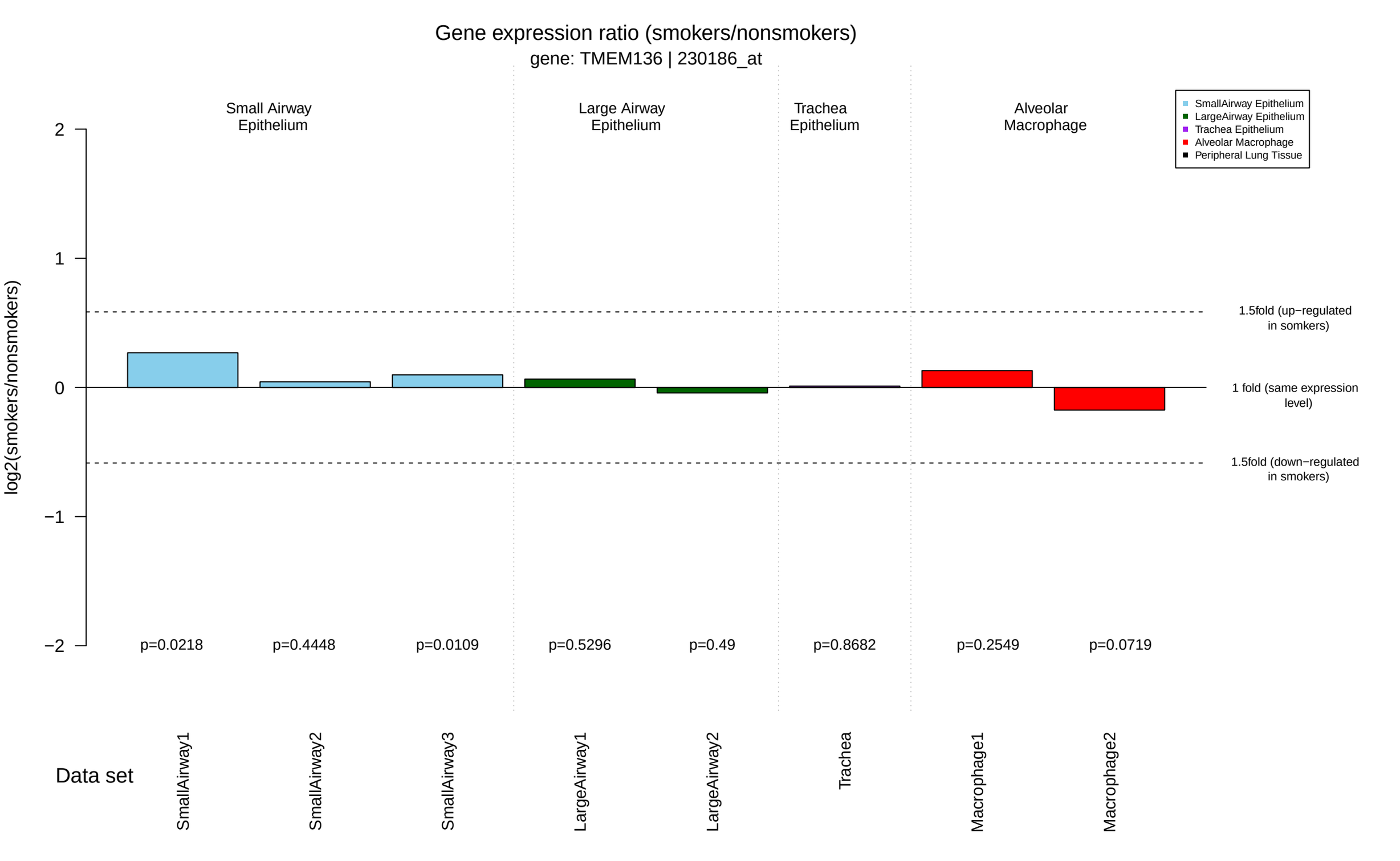
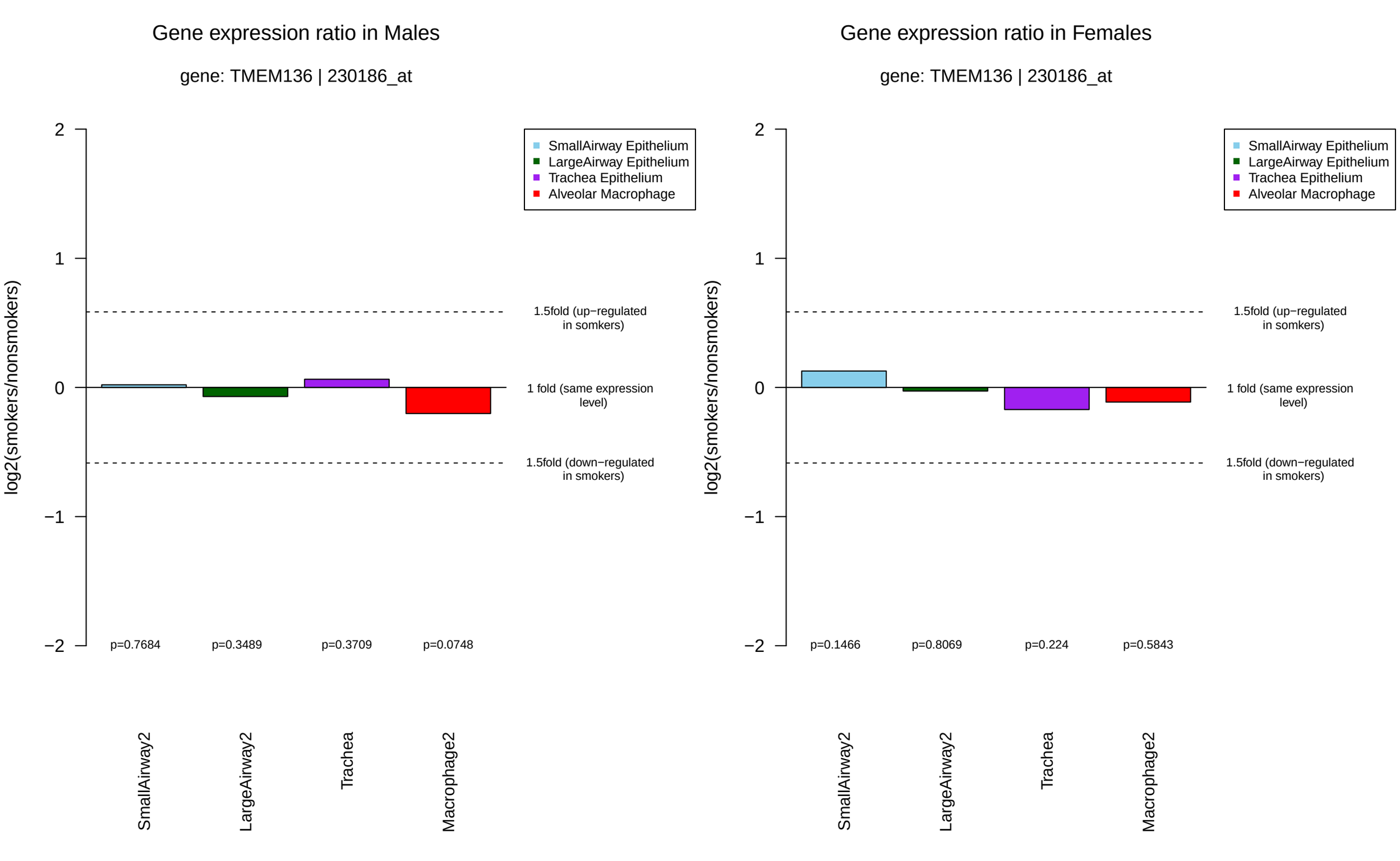
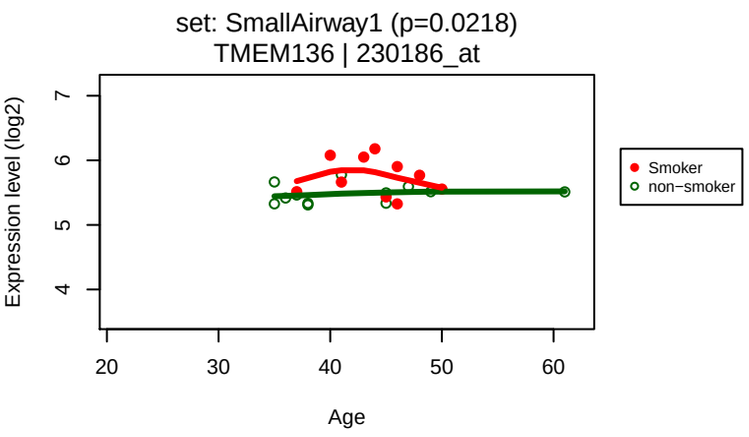
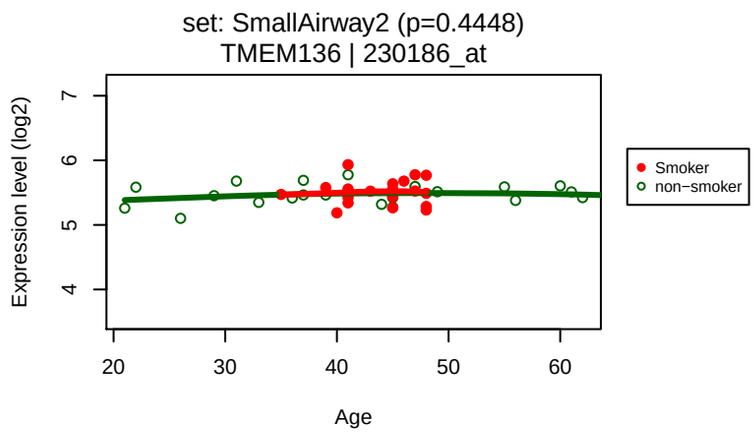
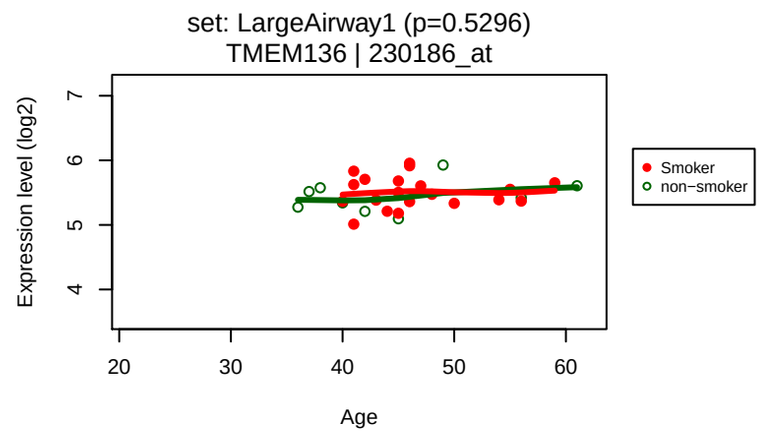

Gene expression of smokers and nonsmokers
gene: TMEM136 | 230186_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0218
p=0.4448
p=0.0109
p=0.5296
p=0.49
p=0.8682
p=0.2549
p=0.0719
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

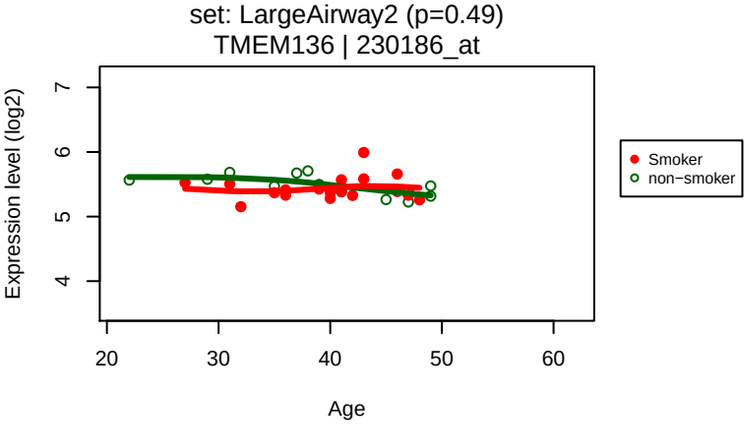

Gene expression ratio (smokers/nonsmokers)
gene: TMEM136 | 230186_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0218
p=0.4448
p=0.0109
p=0.5296
p=0.49
p=0.8682
p=0.2549
p=0.0719
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

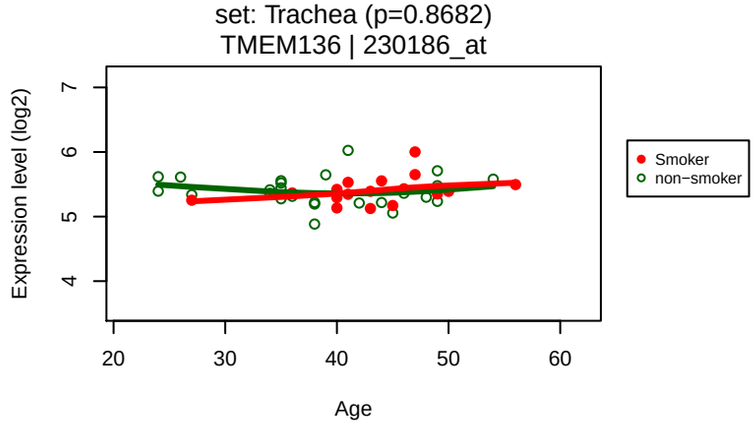

Gene expression ratio in Males
gene: TMEM136 | 230186_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7684
p=0.3489
p=0.3709
p=0.0748
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TMEM136 | 230186_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1466
p=0.8069
p=0.224
p=0.5843
SmallAirway2
LargeAirway2
Trachea
Macrophage2

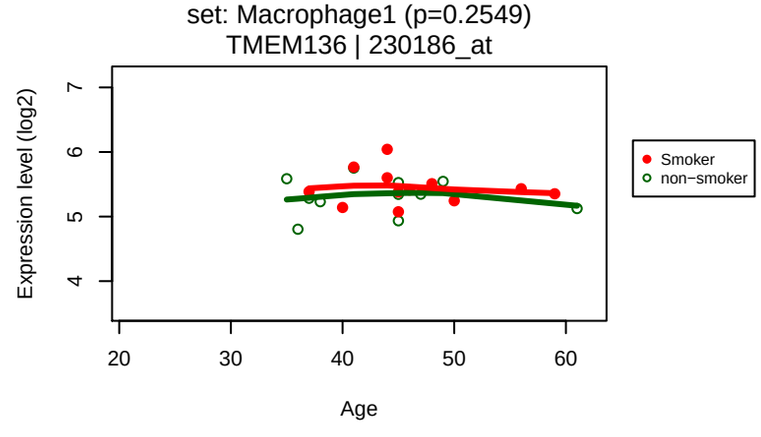

set: SmallAirway1 (p=0.0218)
TMEM136 | 230186_at
Expression level (log2)
Age
Smoker
non−smoker

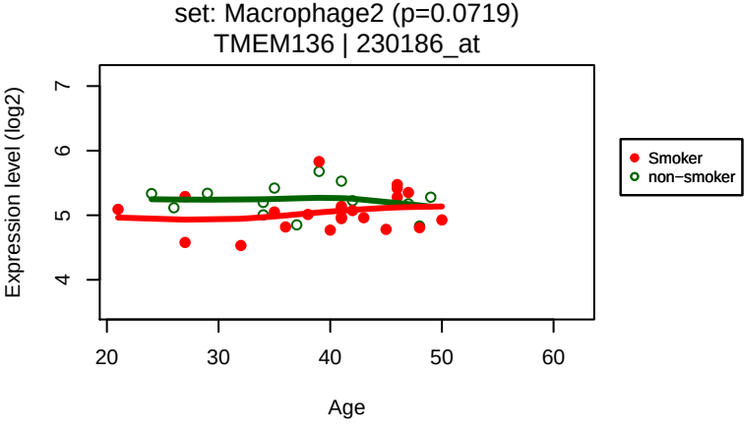

set: SmallAirway2 (p=0.4448)
TMEM136 | 230186_at
Expression level (log2)
Age
Smoker
non−smoker

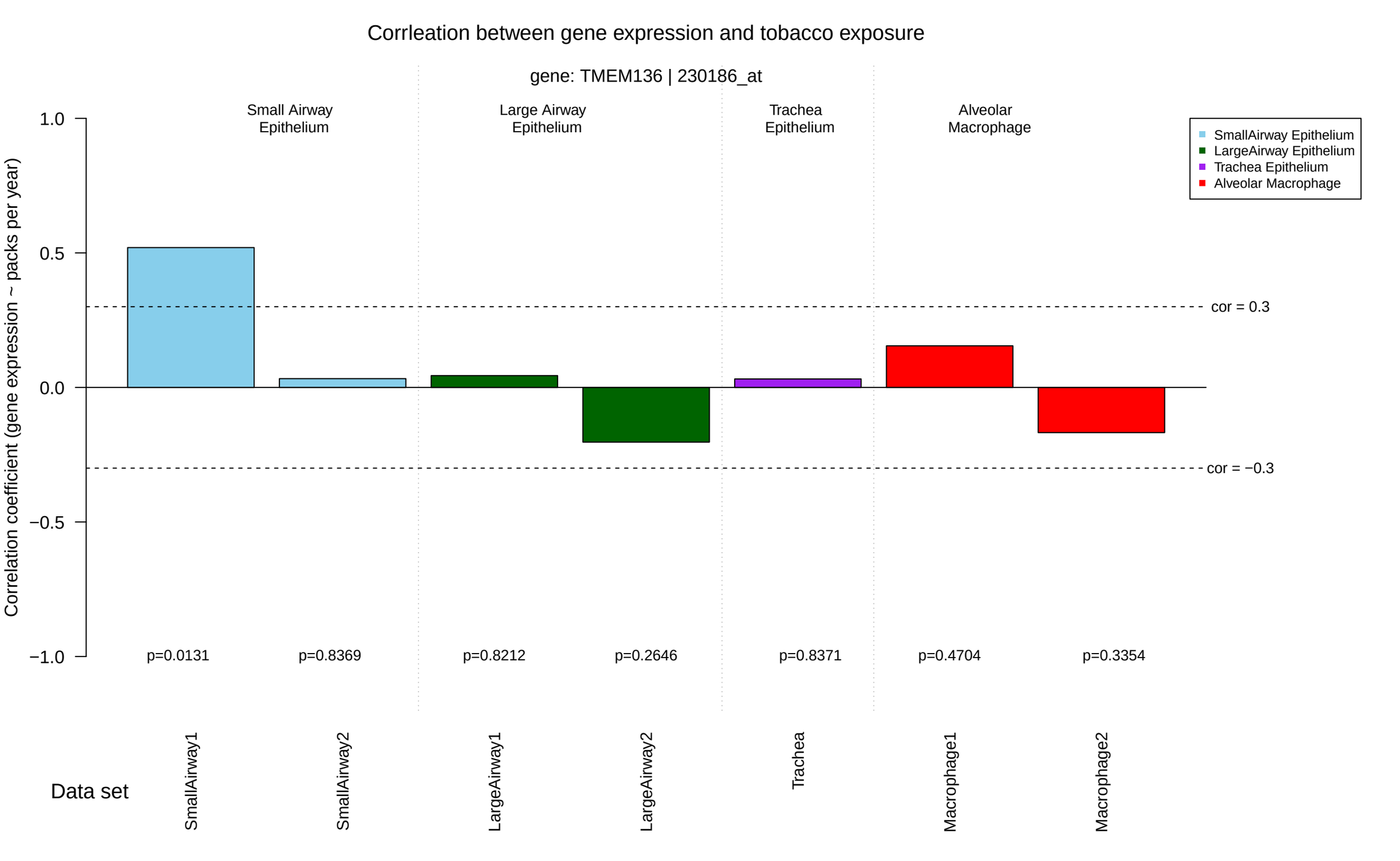

set: LargeAirway1 (p=0.5296)
TMEM136 | 230186_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.49)
TMEM136 | 230186_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8682)
TMEM136 | 230186_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2549)
TMEM136 | 230186_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0719)
TMEM136 | 230186_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TMEM136 | 230186_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0131
p=0.8369
p=0.8212
p=0.2646
p=0.8371
p=0.4704
p=0.3354
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2